Theory Week 3

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Orga

Mechanics 3 CAB G59
WhatsApp group

- Find all doc's on my website (link in Whatsapp bio)
- remember to do the STACK
- MIDTERM (week 8)
  - be in your group
  - counts for bonus (400 pts)
- Study Center

3	30/09/24	work-energy balance, (non-)conservative forces	1.2.3
	02/10/24	single-particle examples, particle impact	1.2.3-1.2.4
		exercise 3: single-particle kinetics	

Work-energy balance (WEB) (p.34-44) Alternative to LMB

Idea: Comparison of two states of a particle (instead entire moment)

... leads to the **work–energy balance** *for a system of particles* as

(p. 59)

$$T(t) = \frac{1}{2}\sum_{i=1}^{n} m_i|\mathbf{v}_i(t)|^2, \quad W_{12} = \sum_{i=1}^{n}\int_{\mathbf{r}_i(t_1)}^{\mathbf{r}_i(t_2)} \mathbf{F}_i \cdot d\mathbf{r}_i \quad \Rightarrow \quad \underbrace{T(t_2) - T(t_1)}_{\text{change in kin. energy}} = W_{12} \qquad (2.23)$$

, where $T(t)$ = kinetic energy

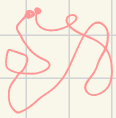
W_{12} = work done on particle between state 1&2 by external forces

Differentiate conservative & non-conservative forces

1. conservative forces

"no loss of energy on closed path"

$E_{start} = E_{end}$



$$W_{12}^{cons.} = \Delta \text{ Potential energy} = V_2 - V_1$$

conservative systems $\Rightarrow$

conservation of energy for a conservative system:

$$T + V = \text{const.}$$

$$T_1 + V_1 = T_2 + V_2$$

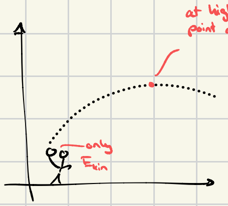
Two main examples

i) Gravity

$T+V = \text{const}$

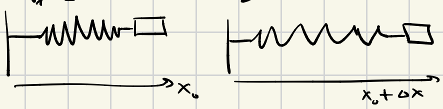
at highest point only E_{pot}

only E_{kin}



ii) Elastic springs

$V_1 = \frac{1}{2}k\Delta x_1^2 = 0$ $\qquad V_2 = \frac{1}{2}k\Delta x_2^2 = \frac{1}{2}k\Delta x^2$



$x_0 \qquad x_0 + \Delta x$

$$V_1 - V_2 = -\frac{1}{2}k\Delta x^2$$

↳ special case of rigid links & systems (we'll get back to this)

- If the link between two particles is **rigid**, one can show that the *internal forces perform no work.* To see this, notice that a rigid link between particles i and j implies

p. 59 & 60

- We can extend the previous point to a **rigid system**: *if a system of particles is rigid, all internal forces perform no work.* This can easily be established by summing over all particles:

we all already know this :)

$$\Rightarrow W_{12}^{cons} = T(t_2) - T(t_1) = V(r_2) - V(r_1) \quad \Rightarrow T(t_1) + V(r_1) = T(t_2) + V(r_2)$$

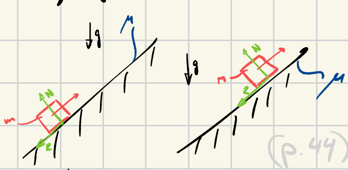
2. non-conservative external forces

"everything other than conservative forces"

$$W_{12} = W_{12}^{non\text{-}cons} + W_{12}^{cons}$$

Main example

i) friction $\qquad$ (ii) drag)



(p. 44)

↳ here R performs work, N doesn't

! Be careful: work is only performed by forces in direction of the trajectory

When do I use WEB?

1. 2 states
   ↳ ratios $\frac{v_1}{v_2}$

2. cons. sys.
   ↳ max-height

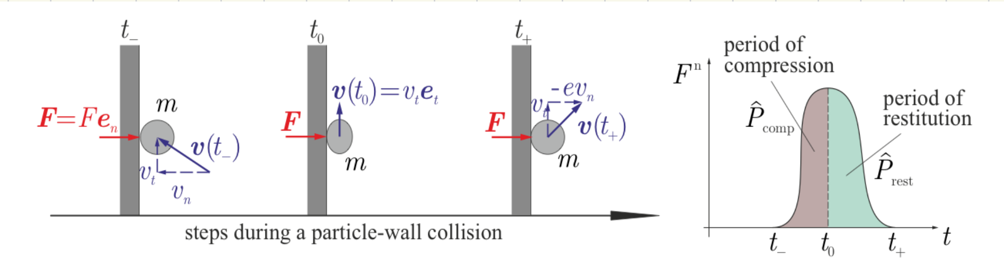
General recipe

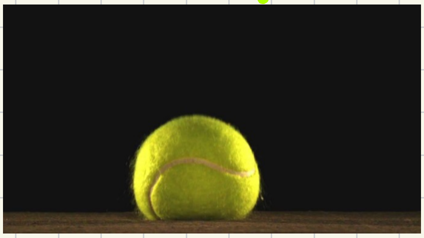
1. choose right reference frame
2. AMB, LMB or WEB?
3. check formula sheet for better inspo if needed

Mostly depend on eachother

Particle impact (p. 45 - 51) (derived from LMB)

Always think how would a tennis ball behave (sanity check)





tangential: $v_t(t_-) = v_t(t_0) = v(t_+)$ → makes sense, since no forces in tan. dir. (1st axiom)

normal: $e \cdot v_n(t_-) = v_n(t_+)$ → e is coeff. of restitution
↳ $e = 1$ ⇒ perfectly elastic → no lost energy ⇒ $T(t_-) = T(t_+)$

in real life: $0 < e < 1$
↳ always some energy loss

1/8 pages of formula sheet done!